Conformal Field Theory

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Chapter 1

Conformal Group

The conformal group is the group of coordinate transformations that preserve angles. This includes translations, Lorentz transformations, and rescalings. Notably, this makes the Poincare group a subgroup of the conformal group.

We can define the conformal group by considering transformations $x \to x'$ that leave the Minkowski metric unchanged up to an overall position dependent rescaling:

$$\eta_{\mu\nu} \to \Omega^2(x)\eta_{\mu\nu}.$$

Theorem: The conformal Killing equation

For any infinitesimal conformal coordinate transformation of the form $x^\mu \to x^\mu + \epsilon^\mu$, ϵ^μ satisfies the following equation:

$$\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu = \frac{2}{d}(\partial \cdot \epsilon)\eta_{\mu\nu}.$$

Proof

Consider an infinitesimal transformation $x^\mu \to x^\mu + \epsilon^\mu$. To leading order in ϵ,

Theorem: Conformal Killing Vectors in $d \geq 3$

The conformal Killing equation in $d \geq 3$ dimensions has four kinds of solutions:

$$\begin{aligned}
\epsilon^\mu(x) &= a^\mu \\
\epsilon^\mu(x) &= m^\mu{}_\nu x^\nu \\
\epsilon^\mu(x) &= \alpha x^\mu \\
\epsilon^\mu(x) &= 2(x \cdot b)x^\mu - b^\mu x^2
\end{aligned}$$

Proof

To-do We are looking for solutions to the equation $\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu = \frac{2}{d}(\partial \cdot \epsilon)\eta_{\mu\nu}$.